



Roche Accelerates Personalized Healthcare R&D While Reducing Costs Using AWS HealthOmics

Learn how life sciences company Roche is shortening research and development cycles using AWS HealthOmics.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Other](#)



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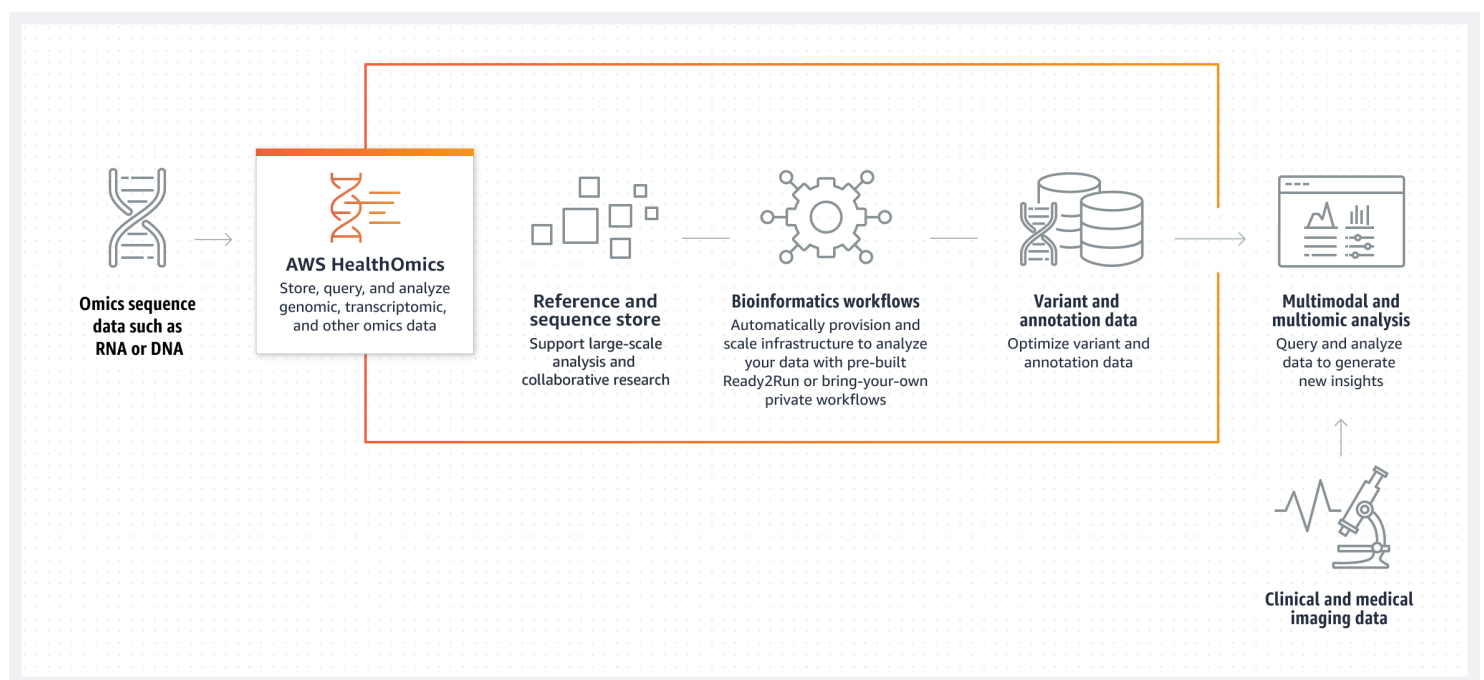
Enhanced security

flexibility, and scalability

Overview

Like many life sciences organizations working to bring better therapeutics to patients, [Roche](#) uses bioinformatics, the combination of biomedical research with mathematical and computational sciences. This burgeoning field is contributing to the rapid evolution of research and development (R&D) in healthcare.

Central to Roche's strategy is its use of Amazon Web Services (AWS) and specifically [AWS HealthOmics](#)—a service that transforms genomic, transcriptomic, and other omics data into insights. From the initial stages of research to drug discovery and point of care, unifying analysis across multimodal and multi-omics health data can lead to powerful insights. Using AWS HealthOmics, Roche's bioinformaticians innovate faster, and patients benefit sooner from personalized medicines.



Opportunity | Improving Genomics Research Using AWS HealthOmics

Founded in Switzerland in 1896, Roche combines pharmaceuticals and diagnostics to advance personalized healthcare and improve lives. Roche strives to offer the right treatments to the right patients at the right time. Its mission requires ongoing clinical research and data-driven medical insights that are supported by robust technology to manage a range of multimodal data from disparate sources.

Roche's IT team created its genomics platform, Research Enhanced Data Commons and Analytical Platform (REDCAP), to help its bioinformaticians efficiently analyze multimodal health data ingested from clinical collaborations and clinical research organizations. However, in its early stages, Roche's on-premise



and compute ran into scaling challenges, required several hours to run workflows, and consumed expensive compute resources.

When AWS HealthOmics launched, Roche saw an opportunity to better support the large-scale analysis of multimodal data. “When we got access to the AWS HealthOmics, we knew that it was going to be a crucial part of our road map,” says Victor Sankarlingam, senior infrastructure architect at Roche. “We no longer needed to estimate our compute capacity. We can also improve data integration and lower storage costs for omics data.”

Solution | Reducing R&D Analysis Time by up to 80%

After an initial experimentation period, Roche began using AWS HealthOmics for its production pipelines shortly after the service launched. It adopted Nextflow and Workflow Description Language as the workflow languages to run its whole-genome sequencing pipelines. Thus, the company has enhanced the scalability and speed of its R&D processes. Bioinformaticians can now navigate standard tasks in the cloud quicker and simpler because AWS HealthOmics automatically selects the optimal compute resources needed to run workflows. Additionally, Roche has incorporated private workflows and AWS HealthOmics [Ready2Run workflows](#), which researchers securely use as needed.

Using AWS HealthOmics, Roche has reduced the time that researchers need to analyze their samples by up to 80 percent. Researchers securely access the platform from across the globe and no longer need to compete for the same resources, so multiple projects can be completed simultaneously. Rather than managing underlying technology or performing lengthy tests or proofs of concept, bioinformaticians focus on their research. “Our bioinformaticians no longer need hand-holding,” says Sankarlingam. “For example, we use AWS HealthOmics to cut the time to complete analysis for cancer research from 1 year to 3 months. That’s the value that we’re getting right now.”

Using AWS HealthOmics, Roche supports its research efforts by facilitating stronger security under the [AWS Shared Responsibility Model](#). The company benefits from secure connections between its data sources and the cloud. These connections are bolstered by multi-factor authentication that meets stringent security requirements. Users can access data from anywhere, at any time, with secure, encrypted connections. They can also manage data access and submit jobs without transferring sensitive data externally. Because all data is protected on AWS, Roche can create a more secure global research environment for developing new therapies in healthcare.

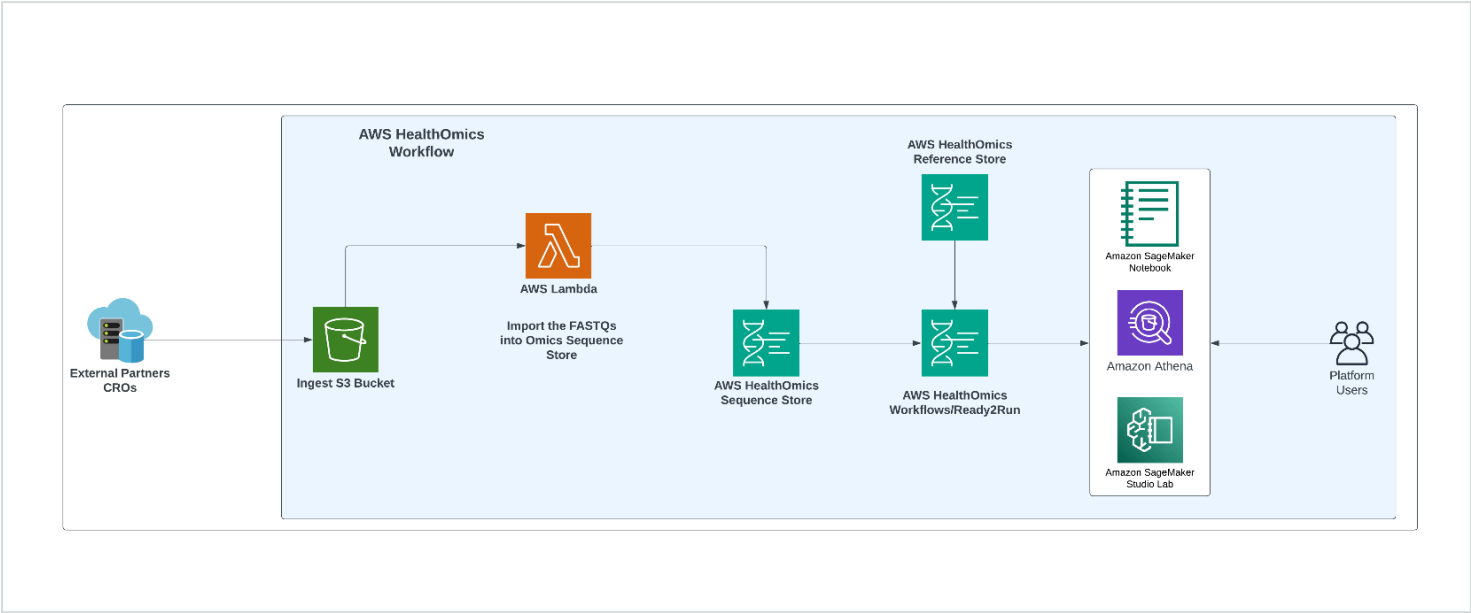


Flexibility is at the core of Roche's use of AWS HealthOmics. Bioinformaticians conduct experiments and analyze research data as needed, with ready access to a suite of customization options and compute resources that emphasize deeper collaboration and R&D efficiency. Roche consistently meets its research needs without having to manage and upgrade physical hardware. Bioinformaticians operate autonomously, initiating and managing workflows and accelerating the pace of discovery in personalized medicine.

As the data used for bioinformatics grows exponentially, Roche can better manage costs. It has saved 40 percent on compute costs by removing idle compute capacity because AWS HealthOmics decommissions compute after a job completes. For storage and archival backup, Roche uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service, to provide employees with a lifecycle policy that automatically archives data after a predetermined period. This has reduced storage costs by 90 percent compared with on-premises methods.

Using purpose-built AWS managed services, Roche has expanded its capabilities beyond genomic analysis and secondary and tertiary analysis. It uses [AWS HealthImaging](#), which lets organizations store, analyze, and share medical images in the cloud at petabyte scale. "Today, we are doing multimodal analytics, which includes image inferencing, image analysis, and image storage," says Sankarlingam. "This would not have been possible on premises."

Architecture Diagram



Outcome | Improving Patient Outcomes with New and Valuable Insights in the Cloud

On AWS, Roche has streamlined the integration and ingestion of multimodal data, positioning itself for success in external collaborations. It can now quickly access and share crucial research data and has established a direct link between its solution and clinical research organizations.

Roche is expanding its use of AWS HealthOmics to the rest of its research projects and aims to create a solution for managing both external and internal data. The company also plans to create an automated pipeline to minimize manual workflow inputs and to further streamline the R&D process, which will empower bioinformaticians to focus even more closely on optimizing workflows.



By adopting AWS HealthOmics, Roche accelerates research outcomes, advancing its mission of enhancing patient care with personalized medicine. “Dealing with early developmental research, our goal is to do whatever we can to achieve results faster,” says Sankarlingam. “On AWS, we can analyze data faster, which helps us find new and valuable insights to improve patient outcomes.”

About Roche

Roche is a biotech company that combines pharmaceuticals and diagnostics to achieve advances in personalized healthcare and improve people’s lives.

AWS Services Used

AWS HealthOmics

AWS HealthOmics is a purpose-built service that helps healthcare and life science organizations and their software partners store, query, and analyze genomic, transcriptomic, and other omics data and then generate insights from that data to improve health.

[Learn more »](#)

AWS HealthImaging

AWS HealthImaging is designed for builders who develop cloud-native medical imaging applications. HealthImaging ingests data in the DICOM P10 format and provides APIs for low-latency retrieval and purpose-built storage.

[Learn more »](#)

Amazon Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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